

# MethylCentroid Theoretical Foundation

## MethylPipeline

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# 1 Goal

**MethylCentroid** summarises a cohort of methylation samples without storing the full sample  $\times$  position matrix. The centroid stores per-position *sufficient statistics* plus an empirical histogram so that downstream comparison can remain ECDF-based.

## 2 Per-Sample Quantities

For sample  $j$  at genomic position  $i$ , define

$$mC_{ij} \quad \text{methylated count,} \tag{1}$$

$$uC_{ij} \quad \text{unmethylated count,} \tag{2}$$

$$c_{ij} = mC_{ij} + uC_{ij} \quad (\text{coverage}), \tag{3}$$

$$x_{ij} = \frac{mC_{ij}}{c_{ij}} \in [0, 1] \quad (\text{methylation fraction, defined when } c_{ij} > 0). \tag{4}$$

Only samples with  $c_{ij} > 0$  contribute to position  $i$ 's centroid statistics.

## 3 Stored Sufficient Statistics

For each genomic position the centroid stores the seven accumulators shown in Table 1.

| Symbol        | Code name   | Definition               |
|---------------|-------------|--------------------------|
| $N$           | <b>N</b>    | $\sum_j 1$               |
| $S_x$         | <b>Sx</b>   | $\sum_j x_{ij}$          |
| $S_{x^2}$     | <b>Sx2</b>  | $\sum_j x_{ij}^2$        |
| $\Sigma m$    | <b>Sm</b>   | $\sum_j mC_{ij}$         |
| $\Sigma u$    | <b>Su</b>   | $\sum_j uC_{ij}$         |
| $\Sigma c^2$  | <b>Sc2</b>  | $\sum_j c_{ij}^2$        |
| $\Sigma wx^2$ | <b>Swx2</b> | $\sum_j c_{ij} x_{ij}^2$ |

Table 1: Sufficient statistics stored per genomic position.

### 3.1 Derived Quantities

The stored accumulators are enough to recover commonly needed summaries without re-reading sample files.

**Mean methylation fraction.**

$$\hat{\mu} = \frac{S_x}{N}. \quad (5)$$

**Sample variance of methylation fractions.**

$$\hat{\sigma}^2 = \frac{S_{x^2} - S_x^2/N}{\max(N-1, 1)}. \quad (6)$$

**Average counts.**

$$\widehat{mC} = \frac{\Sigma m}{N}, \quad \widehat{uC} = \frac{\Sigma u}{N}. \quad (7)$$

**Method-of-moments Beta parameters (derived, not stored).** When a downstream consumer requests Beta parameters, they are computed from  $\hat{\mu}$  and  $\hat{\sigma}^2$ :

$$\hat{\alpha} = \hat{\mu} \left( \frac{\hat{\mu}(1-\hat{\mu})}{\hat{\sigma}^2} - 1 \right), \quad (8)$$

$$\hat{\beta} = (1-\hat{\mu}) \left( \frac{\hat{\mu}(1-\hat{\mu})}{\hat{\sigma}^2} - 1 \right). \quad (9)$$

These parameters are *not* part of the comparison pipeline; they are available only as derived statistics.

**Heterogeneity statistics.** Sc2 and Swx2 allow coverage-weighted variance decompositions, for example DerSimonian–Laird-style between-sample heterogeneity estimates used by the `dl_heterogeneity` helper in MethylUtils. This is why both fields remain part of the persisted schema even though they are not used in the core centroid mean or variance formulas.

## 4 ECDF Histogram

Each supported centroid must also store a per-position histogram over methylation fractions:

- **bins** — number of histogram bins  $B \geq 1$ .
- **bin\_counts** — array of shape  $(n_{\text{positions}}, B)$ .
- **bin\_edges** — reconstructed in memory as uniform edges on  $[0, 1]$ :  $0, 1/B, 2/B, \dots, 1$ .

The parameter `binned_stats.bins` is **mandatory** and must satisfy  $B \geq 1$ . The canonical default is  $B = 20$ .

For each accumulator update with sample  $j$ , the bin index is

$$b_{ij} = \lfloor x_{ij} \cdot B \rfloor \text{ clamped to } \{0, \dots, B-1\}, \quad (10)$$

and `bin_counts` $[i, b_{ij}]$  is incremented.

The histogram defines an *empirical CDF* at the bin edges. A PCHIP (Piecwise Cubic Hermite Interpolating Polynomial) spline is used to reconstruct a continuous CDF  $F(x)$  and PDF  $f(x) = F'(x)$  for any  $x \in [0, 1]$  when downstream code evaluates overlap or log-likelihood.

## 5 Why ECDF Only

The current contract removes runtime switching among Normal, Beta, Beta-Binomial, and Beta-Mixture comparison modes. The ECDF-only design offers several advantages:

- **Shape preservation.** The empirical histogram captures skewed, multimodal, or otherwise irregular methylation distributions without imposing a parametric family.
- **Contract alignment.** Build-time (MethylCentroid) and compare-time (MethylCentroid-Pair, MethylDetector) both rely on the same data model: sufficient statistics plus empirical histograms.
- **Downstream consistency.** ECDFClassifier (MethylDetector classifier stage) uses the same PCHIP PDF model as the DMP detection stage, so the score that selects DMPs is consistent with the score that classifies new samples.

## 6 Cohort Update Semantics

At the runner level, cohort membership is resolved as

$$\text{effective\_samples} = (\text{samples} \setminus \text{remove\_samples}) \cup \text{add\_samples}. \quad (11)$$

That resolved cohort is built for the target chromosome/context and persisted. The final active cohort is written to:

- HDF5 metadata field `samples_used`,
- sidecar config field `samples` in `{chrom}-{ctx}_config.json`.

## 7 HDF5 Schema

The centroid file stores only `methylation_data` datasets; no separate `binned_stats` group is used.

| Dataset / Attribute                       | Content                                      |
|-------------------------------------------|----------------------------------------------|
| <code>methylation_data/pos</code>         | Genomic positions (uint32)                   |
| <code>methylation_data/tnc</code>         | Trinucleotide context (uint8)                |
| <code>methylation_data/N</code>           | Sample count per position                    |
| <code>methylation_data/Sx</code>          | Sum of methylation fractions                 |
| <code>methylation_data/Sx2</code>         | Sum of squared fractions                     |
| <code>methylation_data/Sm</code>          | Sum of methylated counts                     |
| <code>methylation_data/Su</code>          | Sum of unmethylated counts                   |
| <code>methylation_data/Sc2</code>         | Sum of squared coverages                     |
| <code>methylation_data/Swx2</code>        | Coverage-weighted sum of squared fractions   |
| <code>methylation_data/bin_counts</code>  | ECDF histogram ( $n_{\text{pos}} \times B$ ) |
| <code>methylation_data.attrs[bins]</code> | Scalar bin count $B$                         |

Bin edges are *not* stored; they are derived as uniform edges on  $[0, 1]$  from the stored bin count  $B$ .

## 8 Summary

MethylCentroid is built around four design principles:

1. Compact sufficient statistics  $(N, S_x, S_{x^2}, \Sigma m, \Sigma u, \Sigma c^2, \Sigma wx^2)$ .
2. Required ECDF histogram data (`bin_counts`,  $B \geq 1$ ).
3. Deterministic cohort updates via set arithmetic.
4. One downstream comparison mode: ECDF.